

CRAVE THE WAVE

DESCRIPTION: In this event competitors must demonstrate knowledge and process skills needed to solve problems and answer questions regarding all types of waves and wave motion.

COMPETITORS BRING/NEED:

- Writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.
- Protractor
- Ruler
- 2 calculators

NUMBER OF PARTICIPANTS: Maximum of 2

IMPOUND: NO

THE COMPETITION:

1. The competition must consist of both hands-on tasks and written questions related to waves.
 - A. 25-50% of the score must be from the practical portion (hands-on tasks)
 - B. 50-75% must be from the theoretical portion (written questions)
2. The event supervisor may provide some mathematical relationships, but the competitors are expected to demonstrate an understanding of the concepts outlined below.
3. The competition must consist of at least one task/question from each of the following areas:
 - A. General wave characteristics (e.g., wavelength, amplitude, frequency, period)
 - B. Wave types (e.g., transverse, longitudinal, surface, torsional)
 - C. Earthquake/seismic waves (e.g., p-waves, s-waves, Rayleigh waves, Love waves, surface waves)
 - D. Illustrating that the sun's energy arrives as radiation with a wide range of wavelengths, including infrared, visible, and ultraviolet, and that white light is made up of a spectrum of many different colors
 - E. Observing and explaining that light and sound waves can be reflected, refracted, and/or absorbed.
 - F. Electromagnetic spectrum (e.g. order, usage, visible light)

SCORING:

- **High score wins.**
- Each question or station will be assigned a predetermined set of points by the event supervisor and communicated to the teams.
- Ties will be broken with pre-determined tie-breaker questions which are communicated to teams.